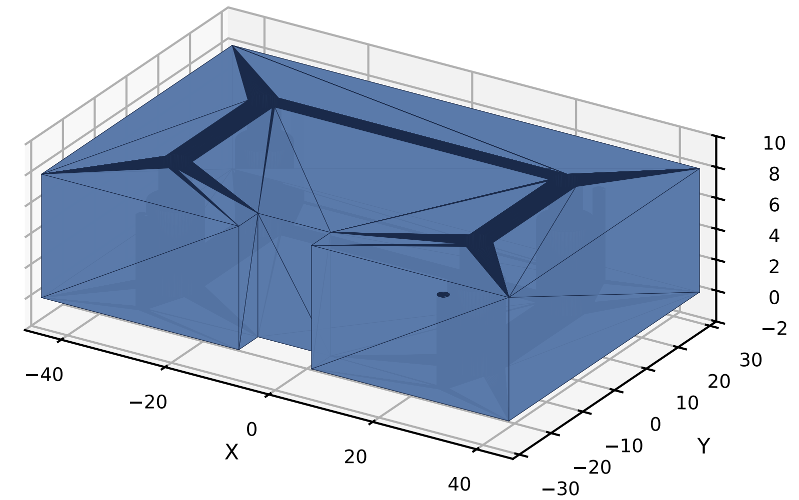
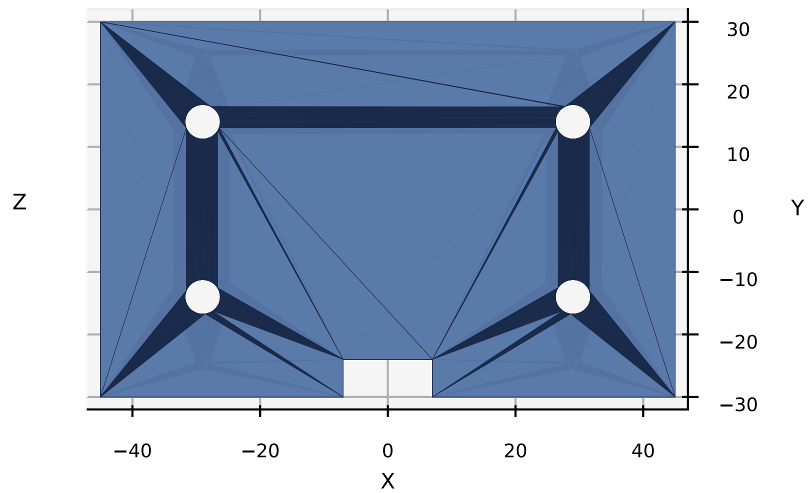
CSM V3 -- HMI Module 10 -- Pi Carrier V1.0

Adapter plate: Cross Beam (58 x 28) -> Raspberry Pi 4 (58 x 49)

Isometric View



Top View (mates beam bottom)

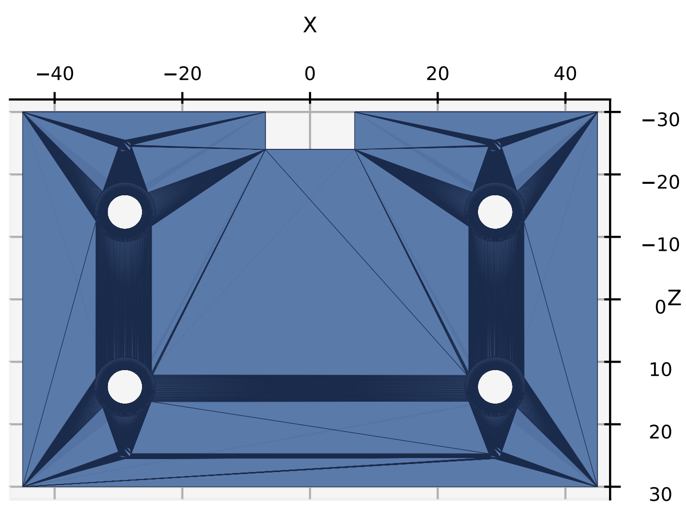


PART	Pi Carrier
NUMBER	M10-P03
VERSION	V1.0 (DRAFT)
MODULE	10 -- HMI
QTY	1
MATERIAL	PETG (PA12 prod)
INFILL	100%, 4 walls
LAYER	0.2 mm
MASS	~52 g
PRINT	flat, ~2 h

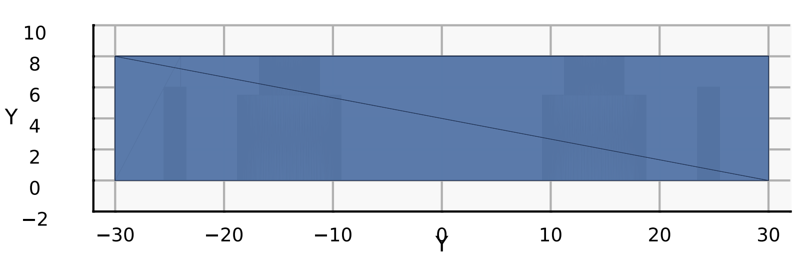
DIMENSIONS (mm)

Plate	90.0 x 60.0 x 8.0
Beam PCD	58.0 x 28.0 (4x M5)
Beam bolt CB	9.5 x 5.5 on bottom
Pi 4 PCD	58.0 x 49.0 (4x M2.5)
Pi pilot	D2.0 x 6.0 (self-tap)
Cable notch	14.0 x 6.0 on -Y edge

Bottom View (4x M5 CB + 4x M2.5 + notch)



Side View (+X) -- counterbore depth



ASSEMBLY

1. Print PETG flat on bed, BOTTOM face down
   100% infill, 4 walls, 0.2 mm layers
2. Heat / press 4x M2.5 standoffs (~10 mm) into
   pilot holes on BOTTOM face (Pi PCD 58 x 49)
3. Mount Raspberry Pi 4 to standoff bottoms
   with M2.5 x 6 mm screws through Pi mount holes
4. Position Pi+carrier assembly below Cross Beam
   bottom face; align with beam Pi inserts
5. M5 x 16 socket-head bolts go UP from carrier
   BOTTOM counterbores into beam BOTTOM inserts
6. Tighten gently -- snug, do not crush plate

INTERFACE CONTRACT (locked):
  Beam PCD 58 x 28 -- HMI_PI_MOUNT_PCD_*
  Pi 4 PCD 58 x 49 -- HMI_PI4_PCD_*

WALL CHECK (all OK):
  edge to beam bolt 11.25 mm
  edge to Pi hole 4.50 mm
  beam to Pi hole 4.75 mm
  cover over CB 2.50 mm